

Date: _____

(Solved Paper) (Spring-2022)

(i)

The Rate of net Investment is $I = 60 \frac{1}{t^3}$ and Capital stock at $t=2$ is 85 find the Capital function K .

$$K = \int I \, dt$$

$$K = \int 60 t^{-1/3} \, dt$$

$$K = 60 \int (t)^{-1/3} \, dt$$

$$K = 60 \frac{(t)^{-1/3 + 1}}{-1/3 + 1} + C$$

$$K = 60 \frac{(t)^{2/3}}{2/3} + C$$

$$K = \frac{3}{1} (60) (t)^{2/3} + C$$

$$K = 3(15) (t)^{2/3} + C$$

$$K = 45 (t)^{2/3} + C \quad \text{G.S}$$

So $t=2$ at $K=85$

$$85 = 45 (1)^{2/3} + C$$

$$85 = 45 + c$$

$$85 - 45 = c$$

by 85

$$c = 85 - 45$$

$$\boxed{c = 40}$$

put (c) in General Solution-

$$\boxed{K = 45(t)^{4/3} + 40}$$

(ii)

Find Integral of

$$y = \int (2x^5 - 3x^{-1/4}) dx$$

given the Initial Conditions
 $y = 6$ when $x = 0$

$$y = \int 2x^5 - \int 3x^{-1/4} dx$$

$$y = 2 \int x^5 - 3 \int x^{-1/4} dx$$

$$y = 2 \frac{x^6}{6} - 3 \frac{(x)^{-1/4+1}}{-1/4+1} + c$$

$$y = \frac{1}{3} x^6 + \frac{4}{3} (x)^{3/4} + c$$

$$y = \frac{1}{3} x^6 + 4(x)^{3/4} + C$$

Now we can apply
initial condition $y = 6$
when x is (0)

$$6 = \frac{1}{3} (0)^6 + 4(0)^{3/4} + C$$

$$6 = C$$

$$C = 6$$

Now put $C = 6$ is
General Solution -

$$y = \frac{1}{3} (x)^6 + 4(x)^{3/4} + 6$$

(iii)

At the certain factory
the MC is $3(q - 4)^2$
dollars per unit when the
level of production is q .
units. By How much will
the Total Manufacturing Cost

$$y = \frac{1}{3} x^6 + 4(x)^{3/4} + C$$

Now we can apply
Initial Condition $y = 6$
when x is (0)

$$6 = \frac{1}{3} (0)^6 + 4(0)^{3/4} + C$$

$$6 = C$$

$$C = 6$$

Now put $C = 6$ is
General Solution -

$$y = \frac{1}{3} (x)^6 + 4(x)^{3/4} + 6$$

(iii)

At the certain factory
the MC is $3(q - 4)^2$
dollar per unit when the
level of production is q .

units. By How much will
the Total Manufacturing Cost

Increasing if the level of production is raised 6 units to 10 units.

$$MC = 3(q - 4)^2$$

So

$$TC = \int MC \, dq$$

Integrate:-

$$\int MC = \int 3(q - 4)^2 \, dq$$

$$TC = \frac{3(q - 4)^3}{3} + C$$

$$\boxed{TC = (q - 4)^3 + C} \quad \text{G.S}$$

Now we need to find change in TC when production raised from 6 to 10 Units

$$\Delta TC = TC(10) - TC(6)$$

So put is G.S

$$TC = (q - 4)^3 + C$$

$$TC = (10 - 4)^3 - (6 - 4)^3 + C$$

$$TC = (6)^3 - (2)^3$$

$$TC = 216 - 8$$

$$TC = 208$$

So the total Manufacturing Cost will increase by 208 Dollars if the level of production is raised from 6 to 10 Units.

(iv) Solve the differential Equations.

$$(x^2 - 3y^2) dx + 2xy dy = 0$$

Dividing both side (x)

$$= \left(\frac{x^2 - 3y^2}{x} \right) dx + \left(\frac{2xy}{x} \right) dy = 0$$

$$= \left(\frac{x^2}{x} - \frac{3y^2}{x} \right) dx + (2y) dy = 0$$

$$= \left(x - \frac{3y^2}{x} \right) dx + 2y dy = 0 \quad \text{--- (i)}$$

Let $\frac{y^2}{x} = u$ Now
Differentiating $^n(u)$ we get
 $du = \frac{1}{x} dy - \frac{y}{x} dx$

(v)

Find the General Solutions

$$y_t'' - 5y_t' + 4y_t = 2$$

for y_p $|y_t = y_c + y_p|$

$$\frac{b}{a_2} = \frac{2}{4} = \frac{1}{2}$$

for y_c

Use D.R.R.c

Case I

$$a_1^2 > 4a_2$$

$$(-5)^2 > 4(4)$$

$$25 > 16$$

So

$$y_c = A_1 e^{r_1 t} + A_2 e^{r_2 t}$$

For r_1, r_2

By Factorization

$$r^2 - 5r + 4 = 0$$

$$r^2 - 4r - r + 4 = 0$$

$$r(r-4) - 1(r-4) = 0$$

$$(r-4)(r-1) = 0$$

$$r-4=0$$

$$r-1=0$$

$$r_2 = -1$$

$$Y_c = A_1 e^{4t} + A_2 e^{2(t)}$$

$$Y_c = A_1 e^{4t} + A_2 e^t \quad \text{--- (i)}$$

For General Solutions

$$Y_t = Y_c + Y_p$$

$$Y_t = A_1 e^{4t} + A_2 e^t + 1/2$$

Gr. Solutions

(vi) Find the Integrations Factor of the Following Equations

$$(7y + 4t^2) dy + 4ty dt = 0$$

$$(7y + 4t^2) dy + 4ty dt = 0$$

$$(7y + 4t^2) dy = -4ty dt$$

$$\frac{1}{y} dy = \frac{-4t dt}{(7y + 4t^2)}$$

$$\frac{1}{y} dy = \frac{-4t}{7y + 4t^2} dt$$

LONG Questions.

(i) Find the difference Equation

$$Y_{t+1} - 0.8Y_t = 0$$

If $Y_2 = 80$ Find Particular Solution Evaluate Y_1, Y_{20}

& Y_{50} .

$$Y_t = Y_c + Y_p$$

For Y_c

$$Y_c = A(-0.8)^t$$

$$Y_c = A - (-0.8)^t$$

$$Y_c = A(0.8)^t$$

where (Y_2) is 80 so we find the value of A

$$Y_p = 0 \Rightarrow \frac{0}{1+0.8} = 0$$

So our General Solution is

$$Y_t = A(0.8)^t + 0$$

put $t = 2$

$$Y_2 = A(0.8)^2$$

$$\therefore Y_2 = 80$$

$$80 = A(0.64)$$

$$\frac{80}{0.64} = A$$

$$[A = 125]$$

Now put in General Solution

$$Y_t = 125(0.8)^T$$

Q.5

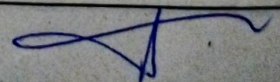
Now we Evaluate

$$Y_1, Y_{20}, Y_{50}$$

$$Y_1 = 125(0.8)^1 = 100$$

$$Y_{20} = 125(0.8)^{20} = 1.441$$

$$Y_{50} = 125(0.8)^{50} = 0.0017$$



(ii) The gross domestic Product (GDP) of a country is \$78 billion and growing at rate of $4.4t^{-1/3}$ billion Dollars Per year after t year. Then use formula to find GDP after 8 years?

Rate change of GDP as

$$\frac{d}{dt} (\text{GDP}) = 4.4t^{-1/3}$$

The initial GDP are given as \$78 billion

$$\text{GDP}(0) = 78$$

• Now find the formula of $\text{GDP}(t)$ we Integrate the rate of change

$$\int \left(\frac{d}{dt} \text{GDP} \right) dt = 4.4 \int (t^{-1/3}) dt$$

$$\text{GDP}(t) = 4.4 \left(\frac{t^{-1/3+1}}{-1/3+1} \right) + C$$

$$GDP_t = 4.4 \left(\frac{3}{2} t^{2/3} \right) + c$$

Now put $t = 0$

$$\therefore GDP_0 = 78$$

$$GDP(0) = 4.4(0) + c$$

$$78 = c$$

$c = 78$, put in General Solution

$$GDP_t = 4.4 \left(\frac{3}{2} t^{2/3} \right) + 78$$

Formula
of $GDP(t)$

Now using this formula

$$GDP_8 = 4.4 \left[\frac{3}{2} (8)^{2/3} \right] + 78$$

$$GDP_8 = 4.4 \left(\frac{3}{2} (4) \right) + 78$$

$$GDP_8 = 4.4 (6) + 78$$

$$GDP_8 = 26.4 + 78$$

$$GDP_8 = 104.4$$

Therefore the GDP after 8 years
approximately \$104.4 billions.

(iii) A Company's sales
rate is $\frac{x}{\sqrt{x+9}}$ scale

per weeks. Find the Formula
for the Total sales after
 x week.

To find the Total sales
after x weeks, we need
to Integral the Term with
respect to x .

$$= \int \frac{x}{\sqrt{x+9}} dx$$

$$= \int \frac{x+9-9}{\sqrt{x+9}} dx$$

$$= \int \frac{x+9}{\sqrt{x+9}} - \int \frac{9}{\sqrt{x+9}} dx$$

$$= \int \sqrt{x+9} dx - 9 \int \frac{1}{\sqrt{x+9}} dx$$

$$= \frac{(x+9)^{1/2+1}}{1/2+1} - 9 \frac{(x+9)^{-1/2+1}}{-1/2+1} + C$$

$$= \frac{2}{3} (x+9)^{3/2} - 9(2) (x+9)^{1/2} + C$$

$$= \frac{2}{3} (n+9)^{3/2} - 18 (n+9)^{1/2} + c$$

Since we are finding the total scale after n weeks we can assume the initial scale at $(n=0)$

$\Rightarrow$ Put $n=0$ in G.S

$$= \frac{2}{3} (0+9)^{3/2} - 18 (0+9)^{1/2} + c$$

$$= \frac{2}{3} (27) - 18 (27) + c$$

$$= 18 - 486 + c$$

$$-468 + c$$

$$\boxed{468 = c}$$

put value of c in General Solution

$$= \frac{2}{3} (n+9)^{3/2} - 18 (n+9)^{1/2} + 468$$

Total Scales